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C4 Diagrams

C4 Diagram: This is an experimental diagram for now. The syntax and properties can change in future releases. Proper documentation will be provided when the syntax is stable.

Mermaid's C4 diagram syntax is compatible with plantUML. See example below:

Code:

mermaid

```
C4Context
```

```
title System Context diagram for Internet Banking System
Enterprise_Boundary(b0, "BankBoundary0") {
    Person(customerA, "Banking Customer A", "A customer of the bank, with a checking account")
    Person(customerB, "Banking Customer B")
    Person_Ext(customerC, "Banking Customer C", "desc")

    Person(customerD, "Banking Customer D", "A customer of the bank, with a savings account")

    System(SystemAA, "Internet Banking System", "Allows customers to perform transactions online")
}

Enterprise_Boundary(b1, "BankBoundary") {
    SystemDb_Ext(SystemE, "Mainframe Banking System", "Stores all of the bank's data")

    System_Boundary(b2, "BankBoundary2") {
        System(SystemA, "Banking System A")
        System(SystemB, "Banking System B", "A system of the bank, with a checking account")
    }

    System_Ext(SystemC, "E-mail system", "The internal Microsoft Exchange system")
    SystemDb(SystemD, "Banking System D Database", "A system of the bank, with a checking account")
}

Boundary(b3, "BankBoundary3", "boundary") {
    System(SystemF, "Banking System F", "A system of the bank, with a checking account")
}
```

```

        SystemQueue(SystemF, "Banking System F Queue", "A system of t
        SystemQueue_Ext(SystemG, "Banking System G Queue", "A system
    }
}
}

```

```

BiRel(customerA, SystemAA, "Uses")
BiRel(SystemAA, SystemE, "Uses")
Rel(SystemAA, SystemC, "Sends e-mails", "SMTP")
Rel(SystemC, customerA, "Sends e-mails to")

```

```

UpdateElementStyle(customerA, $fontColor="red", $bgColor="grey", $b
UpdateRelStyle(customerA, SystemAA, $textColor="blue", $lineColor="
UpdateRelStyle(SystemAA, SystemE, $textColor="blue", $lineColor="bl
UpdateRelStyle(SystemAA, SystemC, $textColor="blue", $lineColor="bl
UpdateRelStyle(SystemC, customerA, $textColor="red", $lineColor="re

```

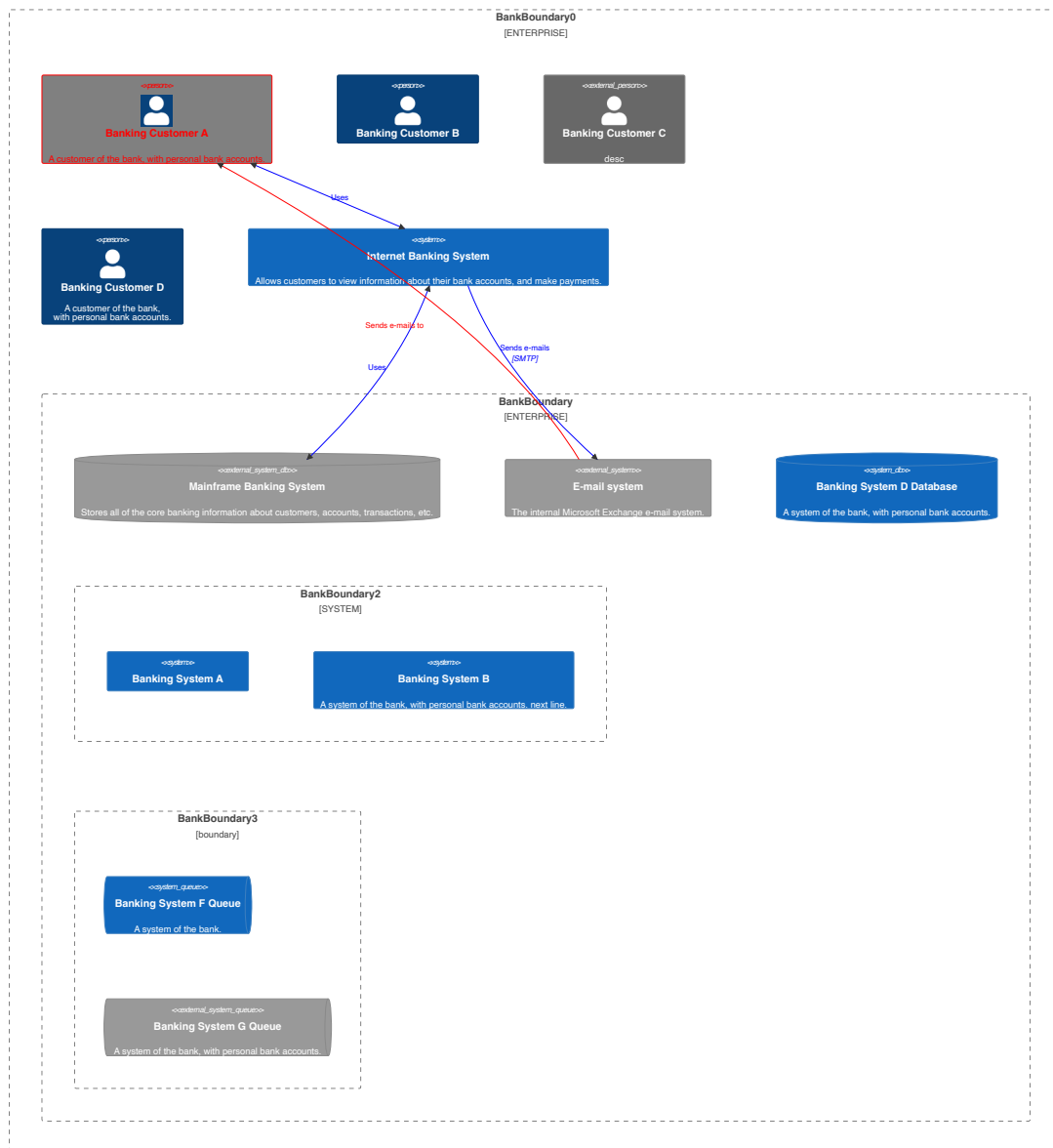
```

UpdateLayoutConfig($c4ShapeInRow="3", $c4BoundaryInRow="1")

```

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System Context diagram for Internet Banking System



For an example, see the source code [demos/index.html](https://mermaid.ai/open-source/syntax/c4.html)

5 types of C4 charts are supported.

- System Context (C4Context)
- Container diagram (C4Container)
- Component diagram (C4Component)
- Dynamic diagram (C4Dynamic)
- Deployment diagram (C4Deployment)

Please refer to the linked document [C4-PlantUML syntax](#) for how to write the C4 diagram.

C4 diagram is fixed style, such as css color, so different css is not provided under different skins. `updateElementStyle` and `UpdateElementStyle` are written in the diagram last part. `updateElementStyle` is inconsistent with the original definition and updates the style of the relationship, including the offset of the text label relative to the original position.

The layout does not use a fully automated layout algorithm. The position of shapes is adjusted by changing the order in which statements are written. So there is no plan to support the following Layout statements. The number of shapes per row and the number of boundaries can be adjusted using `UpdateLayoutConfig`.

- Layout
 - Lay_U, Lay_Up
 - Lay_D, Lay_Down
 - Lay_L, Lay_Left
 - Lay_R, Lay_Right

The following unfinished features are not supported in the short term.

- [] sprite
- [] tags
- [] link
- [] Legend
- [x] System Context
 - [x] Person(alias, label, ?descr, ?sprite, ?tags, \$link)
 - [x] Person_Ext
 - [x] System(alias, label, ?descr, ?sprite, ?tags, \$link)
 - [x] SystemDb
 - [x] SystemQueue
 - [x] System_Ext
 - [x] SystemDb_Ext
 - [x] SystemQueue_Ext
 - [x] Boundary(alias, label, ?type, ?tags, \$link)
 - [x] Enterprise_Boundary(alias, label, ?tags, \$link)

- [x] System_Boundary
- [x] Container diagram
 - [x] Container(alias, label, ?techn, ?descr, ?sprite, ?tags, \$link)
 - [x] ContainerDb
 - [x] ContainerQueue
 - [x] Container_Ext
 - [x] ContainerDb_Ext
 - [x] ContainerQueue_Ext
 - [x] Container_Boundary(alias, label, ?tags, \$link)
- [x] Component diagram
 - [x] Component(alias, label, ?techn, ?descr, ?sprite, ?tags, \$link)
 - [x] ComponentDb
 - [x] ComponentQueue
 - [x] Component_Ext
 - [x] ComponentDb_Ext
 - [x] ComponentQueue_Ext
- [x] Dynamic diagram
 - [x] RelIndex(index, from, to, label, ?tags, \$link)
- [x] Deployment diagram
 - [x] Deployment_Node(alias, label, ?type, ?descr, ?sprite, ?tags, \$link)
 - [x] Node(alias, label, ?type, ?descr, ?sprite, ?tags, \$link): short name of Deployment_Node()
 - [x] Node_L(alias, label, ?type, ?descr, ?sprite, ?tags, \$link): left aligned Node()
 - [x] Node_R(alias, label, ?type, ?descr, ?sprite, ?tags, \$link): right aligned Node()
- [x] Relationship Types
 - [x] Rel(from, to, label, ?techn, ?descr, ?sprite, ?tags, \$link)
 - [x] BiRel (bidirectional relationship)
 - [x] Rel_U, Rel_Up
 - [x] Rel_D, Rel_Down
 - [x] Rel_L, Rel_Left

- [x] Rel_R, Rel_Right
- [x] Rel_Back
- [x] RelIndex * Compatible with C4-PlantUML syntax, but ignores the index parameter. The sequence number is determined by the order in which the rel statements are written.
- [] Custom tags/stereotypes support and skin param updates
 - [] AddElementTag(tagStereo, ?bgColor, ?fontColor, ?borderColor, ?shadowing, ?shape, ?sprite, ?techn, ?legendText, ?legendSprite): Introduces a new element tag. The styles of the tagged elements are updated and the tag is displayed in the calculated legend.
 - [] AddRelTag(tagStereo, ?textColor, ?lineColor, ?lineStyle, ?sprite, ?techn, ?legendText, ?legendSprite): Introduces a new Relationship tag. The styles of the tagged relationships are updated and the tag is displayed in the calculated legend.
 - [x] UpdateElementStyle(elementName, ?bgColor, ?fontColor, ?borderColor, ?shadowing, ?shape, ?sprite, ?techn, ?legendText, ?legendSprite): This call updates the default style of the elements (component, ...) and creates no additional legend entry.
 - [x] UpdateRelStyle(from, to, ?textColor, ?lineColor, ?offsetX, ?offsetY): This call updates the default relationship colors and creates no additional legend entry. Two new parameters, offsetX and offsetY, are added to set the offset of the original position of the text.
 - [] RoundedBoxShape(): This call returns the name of the rounded box shape and can be used as ?shape argument.
 - [] EightSidedShape(): This call returns the name of the eight sided shape and can be used as ?shape argument.
 - [] DashedLine(): This call returns the name of the dashed line and can be used as ?lineStyle argument.
 - [] DottedLine(): This call returns the name of the dotted line and can be used as ?lineStyle argument.
 - [] BoldLine(): This call returns the name of the bold line and can be used as ?lineStyle argument.
 - [x] UpdateLayoutConfig(?c4ShapeInRow, ?c4BoundaryInRow): New. This call updates the default c4ShapeInRow(4) and c4BoundaryInRow(2).

There are two ways to assign parameters with question marks. One uses the non-named parameter assignment method in the order of the parameters, and the other

uses the named parameter assignment method, where the name must start with a \$ symbol.

Example: UpdateRelStyle(from, to, ?textColor, ?lineColor, ?offsetX, ?offsetY)

```
UpdateRelStyle(customerA, bankA, "red", "blue", "-40", "60")
UpdateRelStyle(customerA, bankA, $offsetX="-40", $offsetY="60", $lineColor="red")
UpdateRelStyle(customerA, bankA, $offsetY="60")
```

C4 System Context Diagram (C4Context)

Code:

```
graph TD
    subgraph C4Context
        title System Context diagram for Internet Banking System
        Enterprise_Boundary(b0, "BankBoundary0") {
            Person(customerA, "Banking Customer A", "A customer of the bank, with a checking account")
            Person(customerB, "Banking Customer B")
            Person_Ext(customerC, "Banking Customer C", "desc")

            Person(customerD, "Banking Customer D", "A customer of the bank, with a savings account")

            System(SystemAA, "Internet Banking System", "Allows customers to perform transactions online")

            Enterprise_Boundary(b1, "BankBoundary") {
                System(SystemE, "Mainframe Banking System", "Stores all of the bank's data")

                System_Boundary(b2, "BankBoundary2") {
                    System(SystemA, "Banking System A")
                    System(SystemB, "Banking System B", "A system of the bank, with a checking account")
                }

                System_Ext(SystemC, "E-mail system", "The internal Microsoft Exchange system")
                SystemDb(SystemD, "Banking System D Database", "A system of the bank, with a checking account")
            }

            Boundary(b3, "BankBoundary3", "boundary") {
                SystemQueue(SystemF, "Banking System F Queue", "A system of the bank, with a checking account")
                SystemQueue_Ext(SystemG, "Banking System G Queue", "A system of the bank, with a checking account")
            }
        }
    end
```

```

    }
  }
}

```

```

BiRel(customerA, SystemAA, "Uses")
BiRel(SystemAA, SystemE, "Uses")
Rel(SystemAA, SystemC, "Sends e-mails", "SMTP")
Rel(SystemC, customerA, "Sends e-mails to")

```

```

UpdateElementStyle(customerA, $fontColor="red", $bgColor="grey", $b
UpdateRelStyle(customerA, SystemAA, $textColor="blue", $lineColor="
UpdateRelStyle(SystemAA, SystemE, $textColor="blue", $lineColor="bl
UpdateRelStyle(SystemAA, SystemC, $textColor="blue", $lineColor="bl
UpdateRelStyle(SystemC, customerA, $textColor="red", $lineColor="re

```

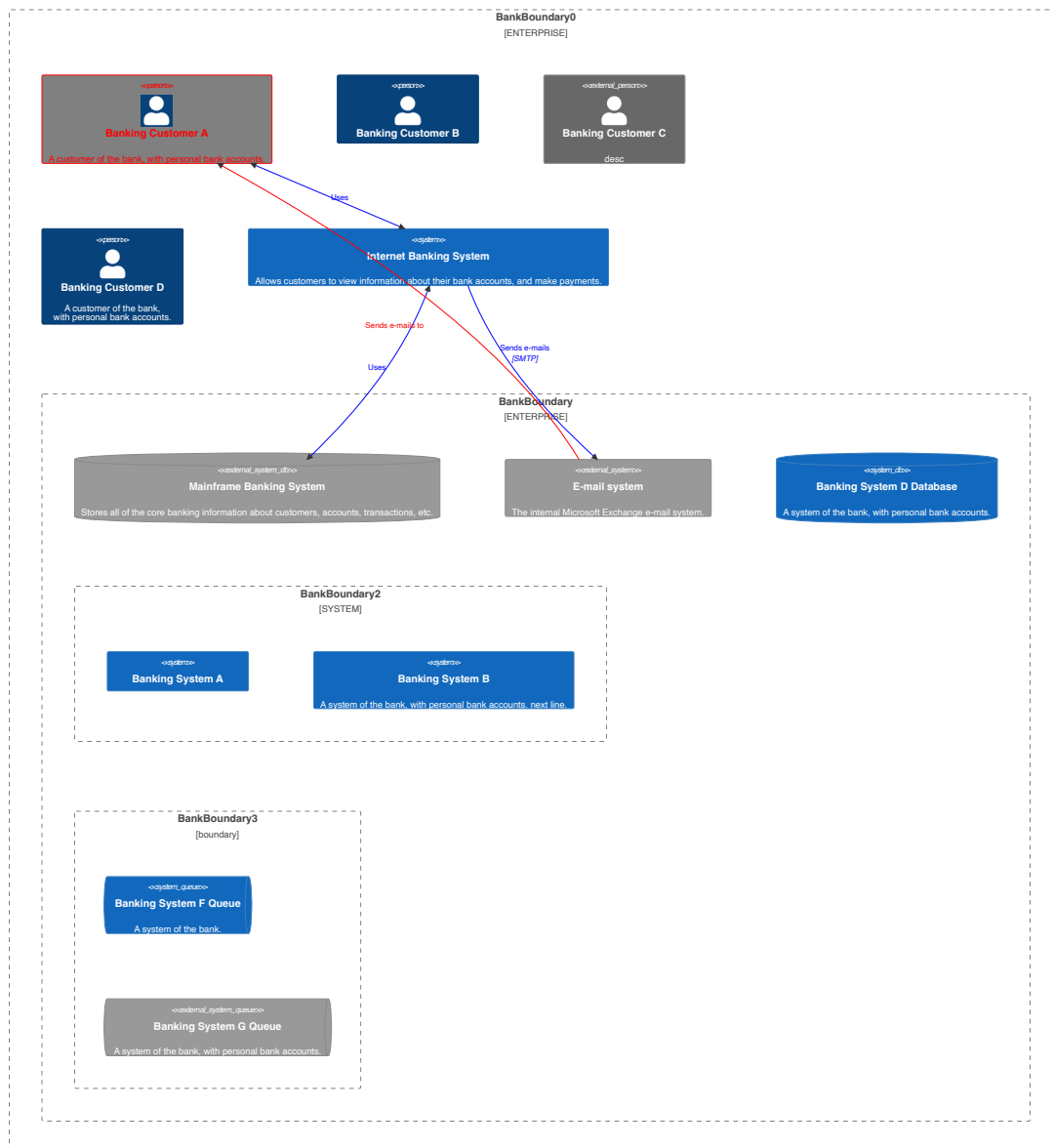
```

UpdateLayoutConfig($c4ShapeInRow="3", $c4BoundaryInRow="1")

```

⌘ + Enter | [Run ▶](#)

System Context diagram for Internet Banking System



C4 Container diagram (C4Container)

Code:

mermaid

```
C4Container
```

```
title Container diagram for Internet Banking System
```

```
System_Ext(email_system, "E-Mail System", "The internal Microsoft Exc
Person(customer, Customer, "A customer of the bank, with personal ban
```

```
Container_Boundary(c1, "Internet Banking") {
```

```

Container(spa, "Single-Page App", "JavaScript, Angular", "Provide
Container_Ext(mobile_app, "Mobile App", "C#, Xamarin", "Provides
Container(web_app, "Web Application", "Java, Spring MVC", "Delive
ContainerDb(database, "Database", "SQL Database", "Stores user re
ContainerDb_Ext(backend_api, "API Application", "Java, Docker Con
}

```

```

System_Ext(banking_system, "Mainframe Banking System", "Stores all of

```

```

Rel(customer, web_app, "Uses", "HTTPS")

```

```

UpdateRelStyle(customer, web_app, $offsetY="60", $offsetX="90")

```

```

Rel(customer, spa, "Uses", "HTTPS")

```

```

UpdateRelStyle(customer, spa, $offsetY="-40")

```

```

Rel(customer, mobile_app, "Uses")

```

```

UpdateRelStyle(customer, mobile_app, $offsetY="-30")

```

```

Rel(web_app, spa, "Delivers")

```

```

UpdateRelStyle(web_app, spa, $offsetX="130")

```

```

Rel(spa, backend_api, "Uses", "async, JSON/HTTPS")

```

```

Rel(mobile_app, backend_api, "Uses", "async, JSON/HTTPS")

```

```

Rel_Back(database, backend_api, "Reads from and writes to", "sync, JD

```

```

Rel(email_system, customer, "Sends e-mails to")

```

```

UpdateRelStyle(email_system, customer, $offsetX="-45")

```

```

Rel(backend_api, email_system, "Sends e-mails using", "sync, SMTP")

```

```

UpdateRelStyle(backend_api, email_system, $offsetY="-60")

```

```

Rel(backend_api, banking_system, "Uses", "sync/async, XML/HTTPS")

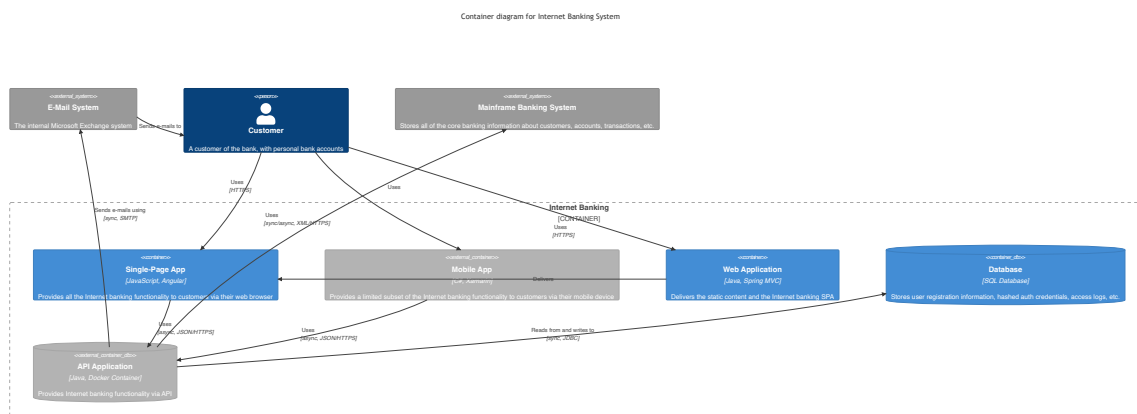
```

```

UpdateRelStyle(backend_api, banking_system, $offsetY="-50", $offsetX=

```

⌘ + Enter | **Run ▶**



C4 Component diagram (C4Component)

Code:

mermaid

```
C4Component
title Component diagram for Internet Banking System – API Application

Container(spa, "Single Page Application", "javascript and angular", "
Container(ma, "Mobile App", "Xamarin", "Provides a limited subset to
ContainerDb(db, "Database", "Relational Database Schema", "Stores use
System_Ext(mbs, "Mainframe Banking System", "Stores all of the core b

Container_Boundary(api, "API Application") {
    Component(sign, "Sign In Controller", "MVC Rest Controller", "All
    Component(accounts, "Accounts Summary Controller", "MVC Rest Cont
    Component(security, "Security Component", "Spring Bean", "Provide
    Component(mbsfacade, "Mainframe Banking System Facade", "Spring B

    Rel(sign, security, "Uses")
    Rel(accounts, mbsfacade, "Uses")
    Rel(security, db, "Read & write to", "JDBC")
    Rel(mbsfacade, mbs, "Uses", "XML/HTTPS")
}

Rel_Back(spa, sign, "Uses", "JSON/HTTPS")
Rel(spa, accounts, "Uses", "JSON/HTTPS")

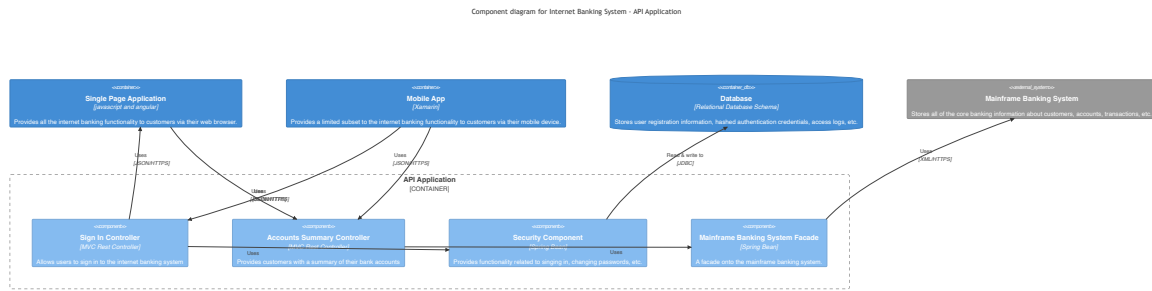
Rel(ma, sign, "Uses", "JSON/HTTPS")
Rel(ma, accounts, "Uses", "JSON/HTTPS")

UpdateRelStyle(spa, sign, $offsetY="-40")
UpdateRelStyle(spa, accounts, $offsetX="40", $offsetY="40")

UpdateRelStyle(ma, sign, $offsetX="-90", $offsetY="40")
UpdateRelStyle(ma, accounts, $offsetY="-40")

UpdateRelStyle(sign, security, $offsetX="-160", $offsetY="10")
UpdateRelStyle(accounts, mbsfacade, $offsetX="140", $offsetY="10")
UpdateRelStyle(security, db, $offsetY="-40")
UpdateRelStyle(mbsfacade, mbs, $offsetY="-40")
```

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C4 Dynamic diagram (C4Dynamic)

Code:

mermaid

C4Dynamic

title Dynamic diagram for Internet Banking System – API Application

```

ContainerDb(c4, "Database", "Relational Database Schema", "Stores user registration information, hashed authentication credentials, access logs, etc")
Container(c1, "Single-Page Application", "JavaScript and Angular", "Provides all the internet banking functionality to customers via their web browser")
Container(c2, "Mobile App", "Swift and Kotlin", "Provides a limited subset to the internet banking functionality to customers via their mobile device")
Container_Boundary(b, "API Application") {
  Component(c3, "Security Component", "Spring Bean", "Provides functionality related to signing in, changing passwords, etc")
  Component(c2, "Sign In Controller", "Spring MVC Rest Controller", "Handles users sign in to the internet banking system")
  Component(c4, "Accounts Summary Controller", "Spring MVC Rest Controller", "Provides customers with a summary of their bank accounts")
}
Rel(c1, c2, "Submits credentials to", "JSON/HTTPS")
Rel(c2, c3, "Calls isAuthenticated() on")
Rel(c3, c4, "select * from users where username = ?", "JDBC")
  
```

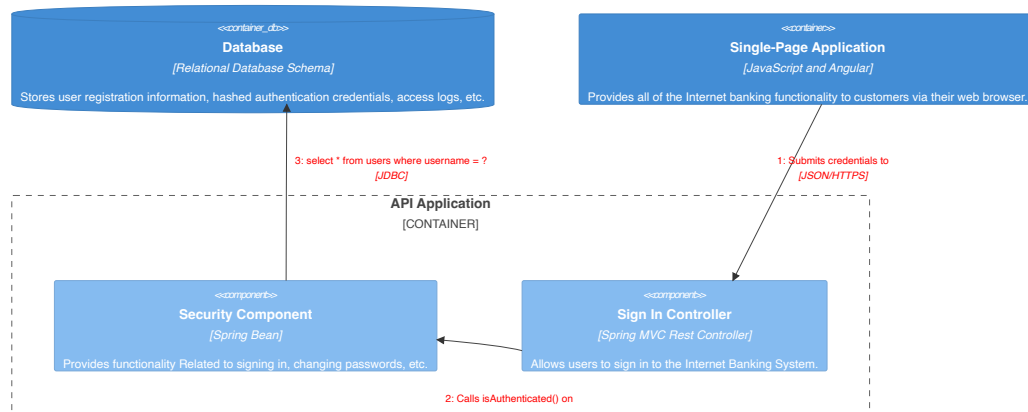
```
UpdateRelStyle(c1, c2, $textColor="red", $offsetY="-40")
```

```
UpdateRelStyle(c2, c3, $textColor="red", $offsetX="-40", $offsetY="60")
```

```
UpdateRelStyle(c3, c4, $textColor="red", $offsetY="-40", $offsetX="10")
```

⌘ + Enter | **Run** ▶

Dynamic diagram for Internet Banking System - API Application



C4 Deployment diagram (C4Deployment)

Code:

mermaid

```

C4Deployment
title Deployment Diagram for Internet Banking System – Live

Deployment_Node(mob, "Customer's mobile device", "Apple IOS or Android")
    Container(mobile, "Mobile App", "Xamarin", "Provides a limited su
}

Deployment_Node(comp, "Customer's computer", "Microsoft Windows or Ap
    Deployment_Node(browser, "Web Browser", "Google Chrome, Mozilla F
        Container(spa, "Single Page Application", "JavaScript and Ang
    }
}

Deployment_Node(plc, "Big Bank plc", "Big Bank plc data center"){
    Deployment_Node(dn, "bigbank-api*** x8", "Ubuntu 16.04 LTS"){
        Deployment_Node/apache, "Apache Tomcat", "Apache Tomcat 8.x")
        Container(api, "API Application", "Java and Spring MVC",
    }
}

Deployment_Node(bb2, "bigbank-web*** x4", "Ubuntu 16.04 LTS"){
    Deployment_Node/apache2, "Apache Tomcat", "Apache Tomcat 8.x"
    Container(web, "Web Application", "Java and Spring MVC",

```

```

    }
  }
  Deployment_Node(bigbankdb01, "bigbank-db01", "Ubuntu 16.04 LTS"){
    Deployment_Node(oracle, "Oracle - Primary", "Oracle 12c"){
      ContainerDb(db, "Database", "Relational Database Schema",
    }
  }
  Deployment_Node(bigbankdb02, "bigbank-db02", "Ubuntu 16.04 LTS")
    Deployment_Node(oracle2, "Oracle - Secondary", "Oracle 12c")
      ContainerDb(db2, "Database", "Relational Database Schema"
    }
  }
}

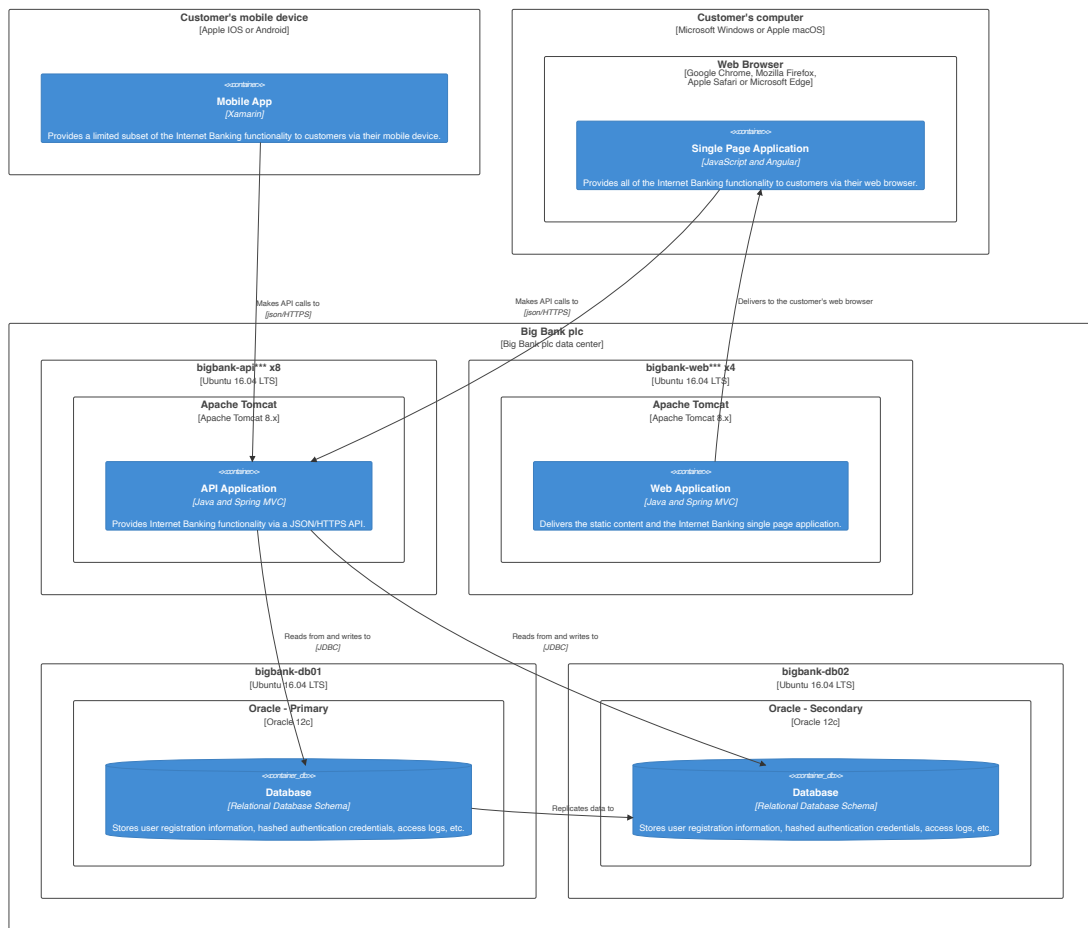
Rel(mobile, api, "Makes API calls to", "json/HTTPS")
Rel(spa, api, "Makes API calls to", "json/HTTPS")
Rel_U(web, spa, "Delivers to the customer's web browser")
Rel(api, db, "Reads from and writes to", "JDBC")
Rel(api, db2, "Reads from and writes to", "JDBC")
Rel_R(db, db2, "Replicates data to")

UpdateRelStyle(spa, api, $offsetY="-40")
UpdateRelStyle(web, spa, $offsetY="-40")
UpdateRelStyle(api, db, $offsetY="-20", $offsetX="5")
UpdateRelStyle(api, db2, $offsetX="-40", $offsetY="-20")
UpdateRelStyle(db, db2, $offsetY="-10")

```

⌘ + Enter | **Run** ▶

Deployment Diagram for Internet Banking System - Live

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